

Tension Reading Device User Manual

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1. Introduction

The Tension Reading Device is a precision instrument designed to measure the force exerted on a string that is mounted on a pulley system, specifically for applications involving small animals, such as rats, in research settings. The system integrates a load cell with an HX711 amplifier, a Raspberry Pi Pico microcontroller, and a custom 3D-printed holder to ensure accurate and reliable tension measurements. Data is displayed and recorded in real-time via a graphical user interface (GUI) on a connected computer, with optional storage on a microSD card. This manual provides detailed instructions for setup, operation, calibration, and maintenance of the device.

2. Safety Precautions

- **Electrical Safety:** Ensure the device is powered off and unplugged before making any connections or adjustments to prevent electrical shock or damage.
 - **Load Cell Handling:** Do not apply excessive force (>100 g) to the load cell, as this may damage the sensor.
 - **Animal Safety:** Ensure the string and attachment mechanism are secure and safe for the rat to prevent injury. Monitor the animal during testing.
 - **Environmental Conditions:** Operate the device in a dry, stable environment ($0\text{--}40^{\circ}\text{C}$, non-condensing humidity) to avoid damage to electronic components.
 - **Proper Setup:** Secure the device firmly to prevent movement during operation, which could skew measurements or harm the test subject.
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3. System Overview

The Tension Reading Device measures the force exerted on a string by a rat using the following components:

- **Load Cell with HX711 Amplifier:** The load cell converts mechanical force into an electrical signal, amplified by the HX711 for high-precision measurements.
- **Raspberry Pi Pico:** A microcontroller that collects data from the load cell and communicates with the computer via USB.
- **Custom 3D-Printed Holder:** A specially designed part that securely mounts the load cell to a 20x20 T-slotted bar, ensuring proper alignment and stability during measurements.
- **Graphical User Interface (GUI):** A Python-based interface for real-time data visualization, unit selection (grams or newtons), taring, and data recording.
- **Optional microSD Card Module:** Allows data to be saved directly to a microSD card for offline analysis.

How the System Works

- The rat pulls on a string attached to the load cell, which is mounted in the 3D-printed holder.
 - The load cell detects the force and generates a proportional electrical signal.
 - The HX711 amplifier converts this signal into a digital format with high resolution.
 - The Raspberry Pi Pico reads the amplified signal at a 13ms interval and sends it to the computer via USB.
 - The GUI displays the force in real-time (updated every 100ms) and allows the user to save data to the computer and/or microSD card.
 - The system automatically tares on startup, and manual taring is available via the GUI to reset the baseline to 0.0g.
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4. Setup

Mechanical Setup

- **Mount the 3D-Printed Holder:**
 - Place the custom 3D-printed holder at the desired location on a 20x20 T-slotted bar.
 - Secure the holder to the bar using two M4 screws. Ensure the holder is firmly attached to prevent movement.
 - The 3D-printed holder is designed to position the load cell such that the string attachment point is aligned with the rat's pulling direction.
- **Attach the Load Cell:**
 - Secure the load cell to the 3D-printed holder using the provided mounting holes and appropriate screws.
 - Ensure the load cell's string attachment point is accessible and unobstructed.
- **Connect the String:**
 - Attach a lightweight, durable string to the load cell's designated attachment point.
 - Ensure the string is securely fastened and aligned with the load cell's sensing axis to avoid off-axis loading, which could affect accuracy.

Electrical Setup

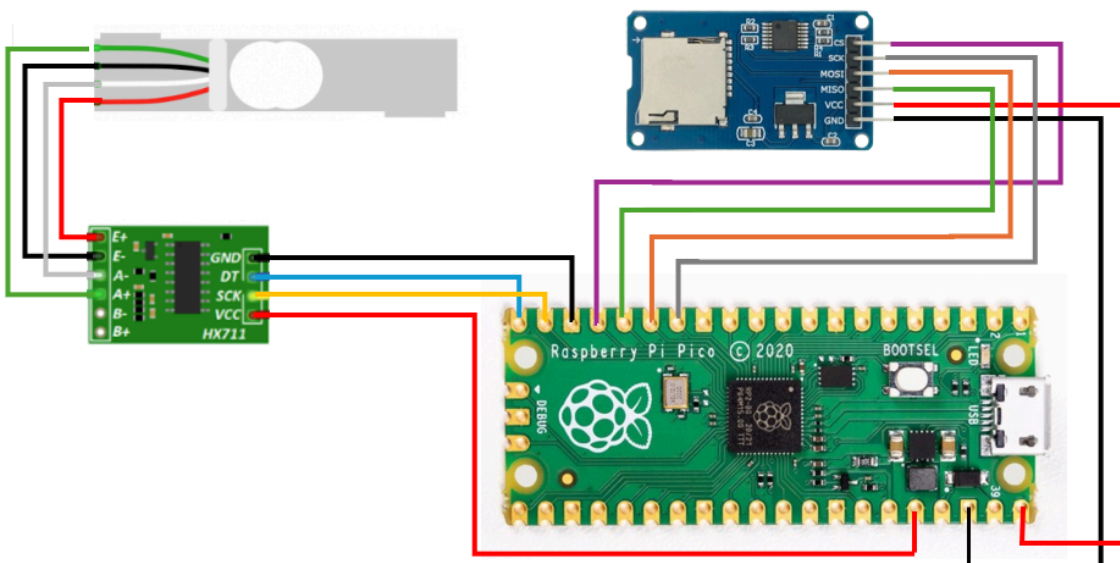
- **Connect the Load Cell to the HX711 Amplifier:** Refer to the wiring diagram in Section 5.
- **Connect the HX711 to the Raspberry Pi Pico:** Use jumper wires to connect the HX711 to the Pico's GPIO pins as shown in Section 5.
- **Optional microSD Card Module:**
 - Plug a microSD card (≤ 32 GB, formatted as FAT32) into the SD card module.
 - Connect the SD card module to the Pico.
- **Power and Computer Connection:**
 - Connect the Raspberry Pi Pico to the computer via a USB cable. The Pico will power on and enter "stand-by" mode, collecting data but not sending it until the GUI is launched.

Software Setup

- **Download the Tension Reader Folder:**
 - Download the Tension Reader folder.
 - This folder contains the GUI code, dependency libraries, and sample data recorded by the load cell.
 - **Install Python:**
 - If Python is not installed, download and install Python 3.x from python.org. See Appendix I for detailed instructions.
 - **Install Dependencies:**
 - Install the `pyserial` and `matplotlib` libraries using pip. See Appendix II for commands.
 - **Verify microSD Card (if used):**
 - Ensure the microSD card is formatted as FAT32 and has sufficient free space. See Appendix IV for formatting instructions.
 - **Ensure “serialPico” port is setup correctly**
 - Edit the [GUI.py](#) file. Ensure that “serialPico” declaration at the start of the code is referencing the correct serial port (set to COM3 by default). See Appendix VII for figuring out which serial port is used.
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5. Wiring Connection Diagram

- The following diagram illustrates the connections between the load cell, HX711 amplifier, Raspberry Pi Pico, and optional microSD card module. Ensure all connections are secure and correct to avoid data errors.



Notes:

- Use jumper wires or a breadboard for connections.
 - Double-check polarity and pin assignments to avoid damaging components.
 - The 3D-printed holder does not affect wiring but ensures the load cell remains stable during measurements.
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6. Operation

- **Power On:**
 - Plug the Raspberry Pi Pico into the computer via USB. The Pico enters “stand-by” mode, collecting data from the load cell but not sending or saving it.
- **Launch the GUI:**
 - Navigate to the Tension Reader folder and run the GUI script (see Appendix V for instructions).
 - The GUI will open, and the Pico will begin sending data, plotting force measurements in real-time (updated every 100ms).
- **Select Units:**
 - Use the “Unit” button to toggle between grams (g) and newtons (N).
- **Record Data:**
 - Click the “Record” button to start saving data.
 - A new folder will be created in the Tension Reader folder on the computer and on the microSD card (if connected).
 - Each time “Record” is turned on within the same session, a new file is created in the same folder.
 - Click “Record” again to stop recording.
- **Stop the Program:**
 - Close the GUI by clicking the “X” button in the top-right corner.
 - All recorded files are saved, and the Pico returns to “stand-by” mode.
- **Disconnect the Raspberry Pi Pico:**
 - Make sure to always disconnect the Pico after a session to avoid crashes and data corruption
 - The Tension Reader is able to run for 2+ hours continuously without issues.

Note: If the GUI fails to open or displays a constant 10.0g reading, refer to the Troubleshooting section.

7. Calibration

- **Automatic Taring:** The Tension Reader automatically tares (sets the baseline to 0.0g) when the Pico is powered on or reset.
 - **Manual Taring:** If a constant load is applied or removed after startup (e.g., the string is adjusted), the baseline may shift. Click the “Tare” button in the GUI to reset the baseline to 0.0g.
 - **Verification:** To ensure accuracy, apply a known weight (e.g., 100g) to the load cell and verify the GUI reading matches the expected value. Contact support if discrepancies persist.
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8. Specifications

- **Measurement Range:** 0–100g (adjustable based on load cell specifications).
 - **Resolution:** Data collected at 13ms intervals by the Pico; displayed at 100ms intervals in the GUI.
 - **Units:** Grams (g) or Newtons (N), user-selectable.
 - **Power Supply:** USB-powered via Raspberry Pi Pico (5V).
 - **Storage:** Optional microSD card (≤ 32 GB, FAT32 format).
 - **Load Cell:** High-precision with HX711 amplifier for 24-bit resolution.
 - **Mounting:** Custom 3D-printed holder for 20x20 T-slotted bar, secured with M4 screws.
 - **Operating Conditions:** 0–40°C, non-condensing humidity.
 - **Git Repository:** <https://github.com/davidpolk/Load-Cell>
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9. Troubleshooting

- **GUI Does Not Open:**
 - Ensure Python, `pyserial`, and `matplotlib` are installed (see Appendices I and II).
 - Verify the Tension Reader folder contains the correct GUI script.
- **Constant 10.0g Reading:**
 - Check load cell and HX711 wiring (see Section 5).
 - Ensure the load cell is not overloaded or damaged.
 - Confirm the Pico is connected to the computer and recognized as a USB device.
 - Verify the correct COM port is selected in the GUI code (see Appendix VII).
 - Reset the Pico by unplugging and replugging the USB cable.
- **Inaccurate Readings:**
 - Perform a manual tare using the GUI “Tare” button.
 - Check for loose connections or off-axis loading on the load cell.
 - Verify the 3D-printed holder is securely mounted.
- **microSD Card Issues:**
 - Ensure the card is $\leq 32\text{GB}$ and formatted as FAT32 (see Appendix IV).
 - Check wiring to the SD card module (see Section 5).

For further assistance, refer to the Contact Information section.

10. Maintenance

- **Cleaning:** Wipe the 3D-printed holder and load cell with a dry or slightly damp cloth. Avoid liquids near electronic components.
- **Load Cell Care:** Inspect the load cell regularly for signs of wear or damage. Replace if accuracy degrades.

- **String Replacement:** Replace the string if it shows signs of fraying or stretching to ensure consistent measurements.
 - **Storage:** Store the device in a dry, dust-free environment when not in use.
 - **Firmware Updates:** Check with the supplier for firmware updates to the Raspberry Pi Pico to improve performance or fix bugs.
 - **3D-Printed Holder:** Inspect for cracks or deformation. Contact the supplier for a replacement if needed.
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11. Contact Information

For technical support, or inquiries:

Email: contact@darionengineering.com

12. Appendices

Appendix I: Installing Python

To install the latest version of python, use the link below

<https://www.python.org/downloads/>

Appendix II: Installing dependencies

Every dependency used by the GUI can be found in /lib inside the GUI folder except for pyserial and matplotlib. These can be installed using pip. To do so, open the terminal on your PC and write the following commands one at a time

- pip install pyserial
- pip install matplotlib

Appendix III: Recording Data

As long as the Pico is powered on, the Tension Reader will constantly be gathering data from the load cell. However, this data will not be sent to the computer, nor saved into files until a command is sent by the GUI (or manually as seen in Appendix VI).

When a recording is started, a new folder will be created within the GUI folder on the computer, and within the SD card if connected. The folder saved in the SD card and Computer will have the same name and hold the exact same data.

In this context, a new “session” is created every time the GUI is opened. As such, only one folder will be created per session. The name of each session folder will be the date-time of when the session has started (when opening the GUI) in the following format: YYYY-MM-DD_HH-MM-SS

Appendix IV: SD Card Specifications

The system works best with a microSD card of 32GB or lower. This is because the SD card must be formatted in FAT32 in order for the Pico to interact with it. To format the SD card:

- Plug SD Card into computer
- Open File explorer and find SD card storage device
- Right-click on our SD Card device then click on Format
- Choose "FAT32" from the *File System* dropdown

Appendix V: Using GUI

There are 3 ways of running the GUI:

- Double click on file
 - To achieve this, you must first right-click on [GUI.py](#)
 - Select Open with... > Python 3.12 (or current python version)
 - Once done this, the user should be able to run the GUI in the future by simply double-clicking on the file
- Run through Terminal
 - Open the terminal on your PC
 - Navigate to the directory where [GUI.py](#) is located
 - Type “python [GUI.py](#)” in the terminal
- Run through Visual Studios or IDE.
 - If you have Visual Studios, VSCode or any other IDE, you can load the code, edit it if needed, and run it through there

- Keep in mind, these IDE's may require Python extensions if not yet installed

Appendix VI: Using Tension Reader without GUI

It is possible to gather and record data without using the GUI. To do so, the user must use an IDE such as Thonny. This is mainly because the user must be able to send commands such as Acquire or Record to the Pico for it to carry out these actions. To do so, the user must:

1. Install Thonny
2. Open Thonny
3. Connect Pico to computer
4. Open communication between Thonny and Pico
 - a. Usually, this can be done by simply clicking on the “STOP” button on Thonny’s top toolbar.
 - b. If that does not work, Click on the bottom right corner of the Thonny window. Normally, it says something like “Local Python 3 - Thonny’s Python” . Then select the serial port corresponding to the Pico (e.g. COM3 Raspberry Pi)
5. Once Thonny has established communication with the Pico, a panel should open on the bottom left corner of Thonny, under the local computer files panel. This panel shows all files stored inside the Pico.
6. Double Click on the “[main.py](#)” file to open it on Thonny
7. Click “Run Current Script” (green play button on Thonny’s top toolbar.)
8. The Pico will start running the program in “Stand-by” mode.
9. To show the printed measurements on the console, type the character **a** (acquire) into Thonny’s console, followed by an *Enter*
10. To start recording (saving data into SD card), type the character **r** (record) into Thonny’s console, followed by an *Enter*
11. The user can use the “Tare” and “Unit” functions by typing **t** and **u** respectively into Thonny’s console, followed by an *Enter*

Appendix VII: Checking Raspberry Pi Pico communication port

In order for the GUI to establish communication with the Pico, it must look at the correct port. By default, the GUI looks at the COM3 port. However, the USB port on your computer might be different. There are two ways of seeing what port the Pico is connected to:

1. Using Device Manager:
 - a. Open the Device manager on your computer
 - b. Scroll down to find and open the “Ports (COM & LPT)” dropdown
 - c. Take note of the COM# connections found in this dropdown
 - d. Connect the Pico to the computer
 - e. At this point, the list will update. A new COM device should appear in the Ports dropdown called “COM#”. This refers to the USB port on which the Pico is connected to.
2. Using Thonny:
 - a. Open Thonny
 - b. Connect Pico to computer
 - c. Open communication between Thonny and Pico
 - i. Usually, this can be done by simply clicking on the “STOP” button on Thonny’s top toolbar.
 - ii. If that does not work, Click on the bottom right corner of the Thonny window. Normally, it says something like “Local Python 3 - Thonny’s Python” . Then select the serial port corresponding to the Pico (e.g. COM3 Raspberry Pi)
 - iii. This “COM#” name that appears next to “Raspberry Pi” in this list refers to the USB port on which the Pico is connected to.

Once you have the name of your COM port, open the [GUI.py](#) file and change the following line:

```
serialPico = serial.Serial("COM3", 115200)
```

so that “COM3” now reflects the USB port on which the Pico is connected to.
